

Sinan Özaydın

PhD, MSc, BS

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BIO

Earth scientist/Geophysicist specialising in magnetotellurics and geodynamic modeling. Sinan's research focuses on understanding tectonic and magmatic processes through interdisciplinary approaches that combine geophysical observations, numerical geodynamic modeling, petrophysical modelling, and petrological-geochemical data. He has developed two scientific software packages (pide, MATE) and contributed to the development of two others (SAnTex, Thermobar).

EMPLOYMENT

Postdoctoral Research Associate at University of Sydney

January 2023 -

Geophysics Consultant at Vox Geophysics

September 2024 -

Postdoctoral Research Fellow at Macquarie University *January 2022 - December 2023*

Postdoctoral Research Associate at Uni. of South Australia *September - December 2021*

EDUCATION

PhD in Geophysics at Macquarie University

2018 - 2021

Sydney, Australia

Thesis Title: *"Three-dimensional magnetotelluric constraints on compositional variations of the Southern African mantle."*

Supervisors: Kate Selway (Primary), William L. Griffin (Associate)

MSc in Geophysics at Boğaziçi University

2015 - 2017

Istanbul, Turkey

Thesis Title: *"The role of crustal fluids in tectonics of north-central Turkey, inferred from three-dimensional magnetotellurics."*

Supervisors: S. Bülent Tank

BS. in Geophysical Engineering at Istanbul Technical University

2009-2015

Istanbul, Turkey

RESEARCH INTERESTS

Geophysics, magnetotellurics, geophysical inversion, petrology, geochemistry, numerical modelling, tectonics, geodynamic modelling, petrophysical interpretation.

PUBLICATIONS

18 - Han, Kui, **Özaydın, Sinan**, Fei, Hongzhan, Man, Lianjie, Wang, Fei, Chanyshv, Artem, Withers, Anthony C., Grayver, Alexander, Katsura, Tomoo (2026). “Lower mantle iron heterogeneity indicated by electrical conductivity”, *Science Advances*, 12(17), [10.1126/sciadv.aec7875](https://doi.org/10.1126/sciadv.aec7875).

17 - **Özaydın, Sinan**, Rey, Patrice F., Chatzaras, Vasileios (2026). “Strain controls the electrical conductivity distribution in the lithosphere”, *Geology*, 54 (2), [10.1130/G53957.1](https://doi.org/10.1130/G53957.1).

16 - Singh, Utpal, **Özaydın, Sinan**, Chatzaras, Vasileios, Rey, Patrice F. (2025) “SAnTex: A Python-based Library for Seismic Anisotropy Calculation”, *Journal of Open Source Software*, 10(110), 6886, [10.21105/joss.06886](https://doi.org/10.21105/joss.06886).

15 - **Özaydın, Sinan**, Li, L., Singh, U., Rey, P.F., Manassero, M.C (2025). “pide: Petrophysical Interpretation tools for geoDynamic Exploration.”, *Journal of Open Source Software*, 10(105), 7021, [10.21105/joss.07021](https://doi.org/10.21105/joss.07021).

14 - Manassero M.C., **Özaydın, Sinan**, Afonso, J.C., Shea, J., Kirkby, A., Ezad, I.S., Thiel, S., Fomin, I., Czarnota, K. (2024). “Lithospheric structure and melting processes in southeast Australia: new constraints from joint probabilistic inversions of 3D magnetotelluric and seismic data”, *Journal of Geophysical Research: Solid Earth*, 129, [10.1029/2023JB028257](https://doi.org/10.1029/2023JB028257).

13 - **Özaydın, Sinan**, Selway, Kate, Foley, Stephen F., Ezad, Isra S., Griffin William L. Tarits, Pascal, Hautot, Sophie (2024). “Role of metasomatism in the development of the East African Rift at the Northern Tanzanian Divergence: Insights from 3D magnetotelluric modelling.”, *Geochemistry, Geophysics, Geosystems*, 25, [10.1029/2023GC011191](https://doi.org/10.1029/2023GC011191).

12 - Selway, Kate, **Özaydın, Sinan**, Payne, Justin (2023). “Metasomatism and depletion of the southern Gawler Craton from combined mantle xenocryst and AusLAMP magnetotelluric data”, *Exploration Geophysics*, 1-15, [10.1080/08123985.2023.2282711](https://doi.org/10.1080/08123985.2023.2282711)

11 - Han, Kui, Guo, Xinzhuang, Wang, Xuben, Zhang, Junfeng, **Özaydın, Sinan**, Li, Dewei, Clark, Simon Martin (2023). “The electrical conductivity of granite: The role of hydrous accessory minerals and the structure water in major minerals.”, *Tectonophysics*, 229857, 856. [10.1016/j.tecto.2023.229857](https://doi.org/10.1016/j.tecto.2023.229857).

10 - Wieser P., Petrelli M., Lubbers J., Wieser E., **Özaydın, Sinan**, Kent A., Till C., (2022). “Thermobar: Open source thermobarometry and hygrometry in Python3.”, *Volcanica*, 5(2), [doi:10.30909/vol.05.02.349384](https://doi.org/10.30909/vol.05.02.349384).

9 - **Özaydın, Sinan**, Selway, Kate (2022). “The Relationship Between Kimberlitic Magmatism and Electrical Conductivity Anomalies in the Mantle.”, *Geophysical Research Letters*, 49, e2022GL099661, [doi:10.1029/2022GL099661](https://doi.org/10.1029/2022GL099661)

8 - **Özaydın, Sinan**, Selway, Kate, Griffin, William L., Moorkamp, Max (2022). “ Probing the southern African lithosphere with magnetotellurics: 2. Linking electrical conductivity, composition, and tectonomagmatic evolution.”, *Journal of Geophysical Research: Solid Earth*, 127, e2021JB023105. [doi:10.1029/2021JB023105](https://doi.org/10.1029/2021JB023105)

7 - Moorkamp, Max, **Özaydın, Sinan**, Selway, Kate, Jones, Alan G., (2022). “Probing the Southern African Lithosphere With Magnetotellurics—Part I: Model Construction”, *Journal of Geophysical Research: Solid Earth*, 127, e2021JB023117. [doi:10.1029/2021JB023117](https://doi.org/10.1029/2021JB023117)

6 - **Özaydın, Sinan**, Selway, Kate, Griffin, William L. (2021). “Are xenoliths from southwestern Kaapvaal Craton representative of the broader mantle? Constraints from magnetotelluric modeling”, *Geophysical Research Letters*, 48, [doi:10.1029/2021GL092570](https://doi.org/10.1029/2021GL092570).

5 - **Özaydın, Sinan**, Selway, Kate, “MATE: An analysis tool for the interpretation of magnetotelluric models of the mantle” (2020). *Geochemistry, Geophysics, Geosystems*, 21, doi:10.1029/2020gc009126.

4 - Selway, Kate, O'Donnell, J. P., **Özaydın, Sinan** (2019). “Upper Mantle Melt Distribution From Petrologically Constrained Magnetotellurics”, *Geochemistry, Geophysics, Geosystems*, 20, doi:10.1029/2019GC008227.

3 - Tank, S. Bülent, **Özaydın, Sinan**, Karas, Mustafa (2018). “Revealing the electrical properties of a gneiss dome using three-dimensional magnetotellurics: Burial and exhumation cycles associated with faulting in Central Anatolia, Turkey”, *Physics of the Earth and Planetary Interiors*, 283, doi:10.1016/j.pepi.2018.07.010.

2 - **Özaydın, Sinan**, Tank, S. Bülent, Karas, Mustafa (2018). “Electrical resistivity structure at the North-Central Turkey inferred from three-dimensional magnetotellurics.”, *Earth, Planets and Space*, 70(49), doi:10.1186/s40623-018-0818-4.

1 - Karas, Mustafa, Tank, S. Bülent, **Özaydın, Sinan** (2017). “Electrical conductivity of a locked fault: investigation of the Ganos segment of the North Anatolian Fault using three-dimensional magnetotellurics.”, *Earth, Planets and Space*, 69(107), doi:10.1186/s40623-017-0695-2.

PUBLICATIONS
IN PROGRESS

Al-Qahtani, Yahya, **Özaydın, Sinan**, Chatzaras, Vasileios, Rey, Patrice F. (2025). “Electrical conductivity structure of Broken Hill and its relationship to Pb-Zn mineralization”, *Journal of Geophysical Research:Solid Earth*, Under Review.

Li, P., Chatzaras, V., Foley, M., **Özaydın, Sinan**, Rey, P. (2026). “A standardized workflow for quantitative 3D shape preferred orientation analysis of minerals using X-ray computed tomography”, *PLOS One*, Under Review.

Rychert, Catherine A. , Dai, Yuhang **Özaydın, Sinan**, Debayle, Eric, Harmon, Nicholas, Chin, Emily J., Conrad, Clinton P., Hirth, Greg, Naif, Samer, Selway, Kate, Artemieva Irina (2026). “Variations and dynamics of the lithosphere-asthenosphere boundary”, *Seismica*, Under Review.

PUBLICATIONS
IN PREPARATION

Singh, Utpal, Chatzaras, Vasileios, **Özaydın, Sinan**, Rey, Patrice F. (2025). “Anisotropic structure beneath North Anatolian Fault.”, *Geophysical Research Letters*, in preparation.

Özaydın, Sinan, Rey, Patrice F., Chatzaras, Vasileios, Hoggard, Mark J. (2025). “Why intraplate earthquakes avoid mantle conductors? A new way to constrain seismic hazard in intra-continental settings.”, *Earth and Planetary Science Letters*, in preparation.

Özaydın, Sinan, Li, Lu, Rey, Patrice F., Chatzaras, Vasileios, Selway, Kate (2026). “A stochastic mantle composition characterization scheme using magnetotellurics and seismic tomography: An application in Southeast Australia.”, *Geophysical Journal International*, in preparation.

Özaydın, Sinan, Munch, Federico, Grayver, Alexander (2026). “Compositional heterogeneity of lithospheric mantle beneath US through petrological-geophysical interpretation”, *To be determined*, in preparation.

REVIEWED MANUSCRIPTS IN	Nature; Geology; Nature Communications; Nature Communications Earth & Environment; Science Advances; Geophysical Journal International; Geophysical Research Letters, Journal of Open Source Software; Gondwana Research; Journal of Geophysical Research: Solid Earth; Tectonics; Geochemistry, Geophysics, Geosystems; Physics of the Earth and Planetary Interiors.
RESEARCH PROJECTS	ARC-DP Identifying mineral systems by geochemical, geophysical and geological mapping of deep Australia 2024 - Research Associate ARC-Linkage Project Evolution of Proterozoic multistage rift basins – key controls on mineral systems 2023 - 2025 Research Associate Compositional constraints on the evolution of Tanzanian Craton and surrounding rift margins using magnetotelluric modelling. 2022 - 2023 Researcher Unveiling the relationship between kimberlites and electrical conductivity signatures of the mantle. 2021 - 2022 Researcher Improving the quantitative interpretations of magnetotelluric models of the cratonic mantle. 2018 - 2022 Researcher / PhD Candidate Summit Station Magnetotellurics (SUMMAT) 2018 Researcher Continental Dynamics / Central Anatolian Tectonics (CD-CAT) 2014-2018 Intern / Researcher Imaging the Shallow Crustal Structure of Ganos Fault by Magnetotellurics 2015-2017 Researcher
CONFERENCE PROCEEDINGS (1 st AUTHOR ONLY)	Özaydın, Sinan (2026). “Geophysical fingerprints of lithospheric heterogeneities.”, University of Cologne. Invited Seminar. Özaydın, Sinan (2026). “Geophysical fingerprints of lithospheric source regions.”, GRC: Geochemistry of Minerals. Oral. Özaydın, Sinan (2026). “Integrating Magnetotelluric and Seismic Observations to Constrain Mantle Composition, Hydration, and Melt Distribution Beneath the Western United States.”, <i>EGU2026</i>, Vienna, Austria. Oral. Özaydın, Sinan (2025). “Thermomechanical modelling as a tool for decoding geophysical anomalies: A case study of a strike-slip pull-apart basin.”, University of Sydney Earthbyte Seminar Series. Invited Seminar. Özaydın, Sinan (2024). “Thermomechanical modelling as a tool for decoding geophysical anomalies,” <i>SGTSG2024</i>, Armidale, Australia. Oral. Özaydın, Sinan (2024) “Building the next generation digital ecosystem for petrophysical inter-

pretation. pide, SAnTex and their applications” University of Tokyo. Invited Seminar.

Özaydın, Sinan (2024). “Entering the anomaly factory: converting geodynamic models into MT models,” *EMIW2024*, Beppu, Japan. Poster.

Özaydın, Sinan (2024). “pide: a python-based petrophysical interpretation tool,” *EMIW2024*, Beppu, Japan. Poster.

Özaydın, Sinan (2023). “Compositional nature and architecture of the lithosphere using magnetotellurics,” University of Sydney Earthbyte Seminar Series. Invited Seminar.

Özaydın, Sinan (2023). “Magnetotelluric insights into mantle rheology and composition,” *AESC2023*, Perth, Australia. Oral.

Özaydın, Sinan (2023). “Compositional architecture of the lithospheric mantle at Northern Tanzanian Divergence,” *AESC2023*, Perth, Australia. Poster.

Özaydın, Sinan (2022). “Compositional causes of electrical conductivity variations in continental lithospheric mantle,” *EMIW2022*, Çeşme, Turkey. Poster.

Özaydın, Sinan (2022). “Lithospheric mantle of Northern Tanzania from 3D magnetotellurics,” *EMIW2022*, Çeşme, Turkey. Poster.

Özaydın, Sinan “Laboratory results coded in MATE,” *EM-Community Webinar Series*, International. **Invited Talk.**

Özaydın, Sinan (2021). “Composition and evolution of the southern African lithosphere,” *AESC2021*, Virtual, Australia. Oral.

Özaydın, Sinan (2020). “Interpretation of conductivity variations with the new software MATE,” *AGU2020*, Virtual, USA. Oral.

Özaydın, Sinan (2019). “3D magnetotelluric models of southern African mantle: hydrogen content and composition,” *AEGC2019*, Perth, Australia. Oral.

Özaydın, Sinan (2018). “Measuring hydrogen content variations in Southern African mantle,” *EMIW2018*, Helsingör, Denmark. Poster.

Özaydın, Sinan (2017). “Deep electrical resistivity structure at Central Pontides, Northern Turkey,” *EGU2017*, Vienna, Austria. Poster.

GRANTS

University of Sydney CAPEX Funding Scheme for MT instruments - 200.000AUD (2023)
NCI Adapter Scheme Q3 - 250 KSU Computing Time at Gadi Supercomputer (2023)
Macquarie University Covid Recovery Postdoctoral Fellowship - Salary + 5000AUD (2021-2022)

AWARDS & ACHIEVEMENTS

Australian Research Council PhD research scholarship 2018-2021
TUBITAK research scholarship 2016-2017

SCIENTIFIC
SOFTWARE

pide - Creator - <https://github.com/sinanozaydin/pide>

A python library to make petrophysical calculations of earth materials from crust to mantle transition zone.

MATE - Creator - <https://github.com/sinanozaydin/MATE>

A software to make quantified interpretations of the magnetotelluric models of the mantle. Published in G-cubed.

SAnTex - Contributor - <https://github.com/utpal-singh/SAnTex>

A python library to calculate elastic properties of rocks.

Thermobar - Contributor - <https://github.com/PennyWieser/Thermobar>

A python library to handle thermobarometric calculations of earth materials.

FIELDWORK
EXPERIENCE

Broken Hill

2024

Installation of BBMT stations around Broken Hill.

Greenland Summit Station Fieldwork

2018

Installation of broad-band and long-period MT stations in Greenland ice sheet.

CD-CAT Fieldwork in Central Anatolia

2014-2018

Installation of nearly 150 broad-band and long-period MT stations in Central Anatolia.

HIGHER DEGREE
RESEARCH
SUPERVISION

Utpal Singh - PhD co-advisor (2023 -), University of Sydney.

Yahya Al-Qahtani - PhD co-advisor (2023 -), University of Sydney.

Puqing Li - PhD co-advisor (2023 -), University of Sydney.